

Teleoperation of Pepper Robot with VR/AR Integration and AI-Driven Insights

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Abstract—This research presents an advanced framework for teleoperating the Pepper humanoid robot, utilizing Virtual Reality (VR), Augmented Reality (AR), Artificial Intelligence (AI), and WebSocket-based real-time communication. Through WebSockets, the system achieves ultra-low latency and bidirectional data exchange, ensuring precise control in dynamic environments. The AI component integrates various models to support autonomous decision-making, while VR provides an immersive interface for direct operator oversight. AR overlays a virtual model of Pepper within the operator's field of vision, delivering continuous feedback on the robot's posture, joint angles, and orientation. This dual-mode control setup, enhanced by AR visualization, optimizes operator interaction and control fidelity, addressing challenges such as latency and scalability. Evaluation demonstrated minimal latency across movement cycles and image processing, with consistent real-time responsiveness critical for high-stakes applications. Image processing latency remained near zero~1ms indicating minimal impact on overall system load, while the overall movement cycle latency showed exceptional stability between 28.0 and 28.2 seconds. AI-driven modules, including depth estimation, facial recognition, and speech recognition, reduced cognitive load and enhanced task adaptability, positioning Pepper as an effective tool for complex teleoperation tasks. These results underscore the system's robustness and effectiveness in elevating real-time operator control and interaction quality, setting new benchmarks in human-robot collaboration.

Keywords—Teleoperation, Humanoid Robot, Artificial Intelligence, Virtual Reality, Augmented Reality

I. INTRODUCTION

Teleoperation—the remote control of robotic systems has evolved remarkably through the integration of VR, AR, and AI. These advancements have transformed teleoperation, enabling robots to perform with unprecedented precision and adaptability in high-stakes environments where direct human presence is impractical

or unsafe. In industrial settings, teleoperation systems with haptic feedback and VR interfaces allow operators to execute intricate tasks with enhanced spatial awareness and control, even across long distances [1]. Underwater exploration, which demands high situational awareness, has also benefited greatly from teleoperation. Digital twin models have been applied to create real-time simulations, allowing operators to navigate and interact with challenging underwater environments more effectively [2]. NASA's Robonaut exemplifies the next level of teleoperated humanoids in space, where VR and AI have been crucial for remote operation under latency constraints, supporting astronauts with maintenance tasks in the harsh conditions of outer space [3]. In healthcare, teleoperated systems are similarly employed for delicate surgeries, allowing physicians to execute intricate procedures remotely, thus reducing risk and improving patient outcomes. A prominent example is the Da Vinci Surgical System [4]. This research extends these advancements by introducing an innovative teleoperation framework for the Pepper humanoid robot, uniquely designed to deliver an immersive, interactive experience for operators. Leveraging VR for intuitive, real-time control, and AR for continuous feedback on Pepper's physical state, the system provides a highly responsive interface that goes beyond conventional teleoperation setups. AI components embedded within the framework assist with tasks requiring autonomy, ensuring seamless transitions between manual and AI-supported control. At the core of this framework is a robust WebSocket-based communication protocol that minimizes latency, facilitating precise, bidirectional data exchange between operator and robot. This configuration not only addresses typical teleoperation challenges, such as response delay and control scalability but also sets a new standard for human-robot collaboration in dynamic environments. By pioneering this integration of VR, AR, and AI, this research advances Pepper's capabilities, showcasing how humanoid robots can enhance human-machine interaction in real-time, laying the groundwork for broader applications of teleoperated humanoids.

II. RELATED WORK

Teleoperation has significantly advanced human-robot interaction by enabling precise control over robotic systems in high-stakes environments. However, challenges like latency, sensor integration, and feedback precision continue to limit these systems. Various studies have addressed these issues. Cerón et al. developed a teleoperation framework using Kinect and Meta Quest for the NAO robot, enhancing interaction for educational and therapeutic applications; however, the system's effectiveness was limited by network latency and Kinect's restricted field of view, affecting real-time synchronization through Robot Operating System (ROS) [5]. Similarly, Zhang's five-dimensional VR teleoperation system for industrial settings combined ROS and Unity to improve control and visualization, achieving fluid interaction but facing scalability limitations due to system complexity [6]. Vicarios platform integrates VR visualization with gesture-based controls for teleoperation, utilizing ROSbridge and TCP protocols over Ethernet for low-latency data exchange. While effective, network latency and the absence of haptic feedback limit real-time performance in complex environments [7]. In other research, an AR-driven teleoperation approach using

HoloLens 2 and TCP/IP protocols enhanced robotic arm control by providing visually guided, intuitive interaction. Although it achieved improved task precision, the AR device's limited field of view and latency issues impacted its performance in more complex tasks [8]. A shared control framework for robotic grasping, which utilized point cloud data and manipulability analysis, effectively combined human input with automation to improve task efficiency, yet struggled with object diversity and varying lighting conditions, which affected grasping precision [9]. Another innovative approach employed VR-SLAM for navigating unknown environments, achieving high grasping accuracy but facing usability issues due to tracking errors in 3D model reconstruction [10]. Additionally, a VR-based ROV teleoperation system for underwater applications integrated a full-body haptic suit for immersive control, enhancing operator precision through synchronized haptic feedback. However, network-induced delays affected real-time control, especially in complex underwater settings [11]. Table 1. summarizes and compares these different teleoperation methods, highlighting their unique methodologies, communication protocols, results, and limitations.

TABLE 1. COMPARISON OF ADVANCED TELEOPERATION METHODS IN HUMAN-ROBOT INTERACTION

Method	Methodology	Communication Methodology	Results	Limitations	References
Kinect & Meta Quest	Captures full-body motion using Kinect sensors and relays data to Meta Quest VR for immersive visual feedback; data processed for joint-based motion mapping.	ROS framework supports real-time communication, synchronizing sensor inputs with the NAO robot's motors and enhancing feedback fidelity.	Highly effective for educational and therapeutic applications, with enhanced interaction fidelity and natural movement mirroring.	Network latency impacts responsiveness, and Kinect's limited field of view reduces precision in spatially complex tasks.	[5]
Five-Dimensional VR System	Implements a five-dimensional VR model to optimize control, visualization, and usability, leveraging ROS and Unity for task versatility.	ROS-Unity synchronization for seamless VR-robot data streaming, providing real-time feedback loops essential for detailed operations.	Significantly improves operator-robot collaboration, especially in complex industrial scenarios requiring detailed task handling.	System complexity limits scalability, making adaptation to broader environments challenging without simplification.	[6]
Vicarios Teleoperation Platform	Uses VR visualization and gesture-based controllers to enable intuitive teleoperation in real-time.	Employs ROSbridge and TCP over Ethernet for low-latency, stable data exchange across the VR interface and robot.	Enhances task accuracy and operator control through a flexible VR viewpoint system, improving success rates.	Limited by network latency and lack of haptic and auditory feedback, impacting complex task responsiveness.	[7]
HoloLens AR System	Uses AR visualization to provide intuitive guidance for teleoperated robotic tasks, overlaying real-time spatial cues for operators.	TCP/IP protocol connects HoloLens AR interface with ROS-enabled robotic systems for synchronized guidance and control.	Increases precision and intuitive control through AR overlays, ideal for tasks requiring visual-spatial alignment.	Field of view limitations in AR device and latency issues in TCP/IP communication affect handling of complex tasks.	[8]
VR-Based Shared Control Framework	Integrates VR with shared control to facilitate complex object grasping using point cloud-based manipulability analysis.	ROS transmits data streams to adjust force and provide stability during grasp, blending human input with automation.	Decreases operator workload while improving grasp stability in multi-object tasks, optimizing task time.	Limited performance with non-symmetrical objects and sensitive to lighting conditions, reducing adaptability.	[9]
SLAM-based VR System	Implements SLAM for dynamic environment mapping and VR model creation to support complex navigational tasks.	Data transmitted via Wi-Fi (ESP8266 module) for synchronization, allowing real-time VR-based spatial guidance.	Reduces operator workload, enabling effective navigation and task execution in unknown or semi-unknown environments.	Requires pre-mapped objects, and 3D tracking errors can occur, reducing effectiveness in unstructured settings.	[10]
VR-Controlled ROV with Haptic Suit	Utilizes full-body VR control with haptic feedback to deliver immersive underwater teleoperation for ROVs.	WebSocket protocol synchronizes VR input with ROV actions in real-time, providing tactile feedback for depth handling.	Enhances operator navigation and control accuracy in underwater environments, with improved spatial awareness.	Network-induced delays impact real-time responsiveness, critical for precise operations in complex underwater scenarios.	[11]

III. PROBLEM STATEMENT

As teleoperation technology advances, achieving precise, real-time control over humanoid robots in complex environments remains a critical challenge. Current systems are hindered by latency issues that disrupt the fluidity of control, causing delays between operator commands and robot responses, which undermines precision in dynamic tasks and reduces overall responsiveness, limiting effectiveness in high-stakes applications. Additionally, inadequate sensor integration compromises real-time feedback and depth perception, making it difficult for operators to navigate intricate environments and execute fine-grained tasks that demand high levels of precision and sensitivity. Scalability further limits current frameworks, as performance tends to degrade under diverse and demanding tasks, restricting adaptability to new operational needs. To address these challenges, this research integrates advanced VR and AR for immersive operator interaction, AI for semi-autonomous decision-making, and a WebSocket-based communication system to enable low-latency, bidirectional data exchange. By reducing latency, enhancing real-time feedback, and optimizing control precision, this framework redefines humanoid robot capabilities and expands human-robot collaboration, making teleoperated robots transformative tools in environments where safety or accessibility limits direct human intervention.

IV. METHODOLOGY

The methodology shown in figure 1. builds on previous studies from Table 1., addressing their limitations to achieve a more effective teleoperation system. To enhance processing efficiency, data exchange between the operator and Pepper is facilitated through a companion processor attached to Pepper. This processor, leveraging the Jetson Orin with 275 TOPS of AI performance, advanced memory capacity, and GPU capabilities, is utilized for tasks requiring substantial processing power, such as running AI

models. Data that requires high computational resources is sent to this processor, while other data bypasses it for direct communication. As shown in Figure 1., the teleoperation framework is organized into several specialized threads, each designed to optimize interaction between the operator and Pepper. The joints & sensors thread continuously collects joint angles and sensor data from Pepper, ensuring that the operator receives accurate, real-time updates on Pepper's physical state. The camera thread captures live video from Pepper's perspective, streaming these visuals to the operator's VR goggles to provide situational awareness and to the AI models for processing. The AI Models then analyze the incoming camera feed to identify objects, estimate depth, or other relevant environmental details, thereby enhancing the operator's situational understanding. Meanwhile, the user interface thread renders an AR model of Pepper, updating it in real-time with sensor data to offer a seamless, immersive experience for the operator. Additionally, the Imitation of the user motion thread interprets the operator's head and hand movements detected by the VR/AR setup, mapping these gestures directly onto Pepper's actions. This enables precise, responsive control, making the robot's movements accurately reflect the operator's intentions. Together, these threads operate in parallel within a modular framework, allowing the system to achieve comprehensive, synchronized teleoperation that balances responsiveness, real-time control, and accuracy across a range of complex tasks. The system's performance will be evaluated using four key metrics image capture latency, image processing latency, movement latency, and overall movement cycle latency.

A. Dual-Protocol Communication for Optimized Teleoperation

Figure 1. illustrates the teleoperation system architecture, highlighting the data pathways for various types of information flow. To ensure optimal data

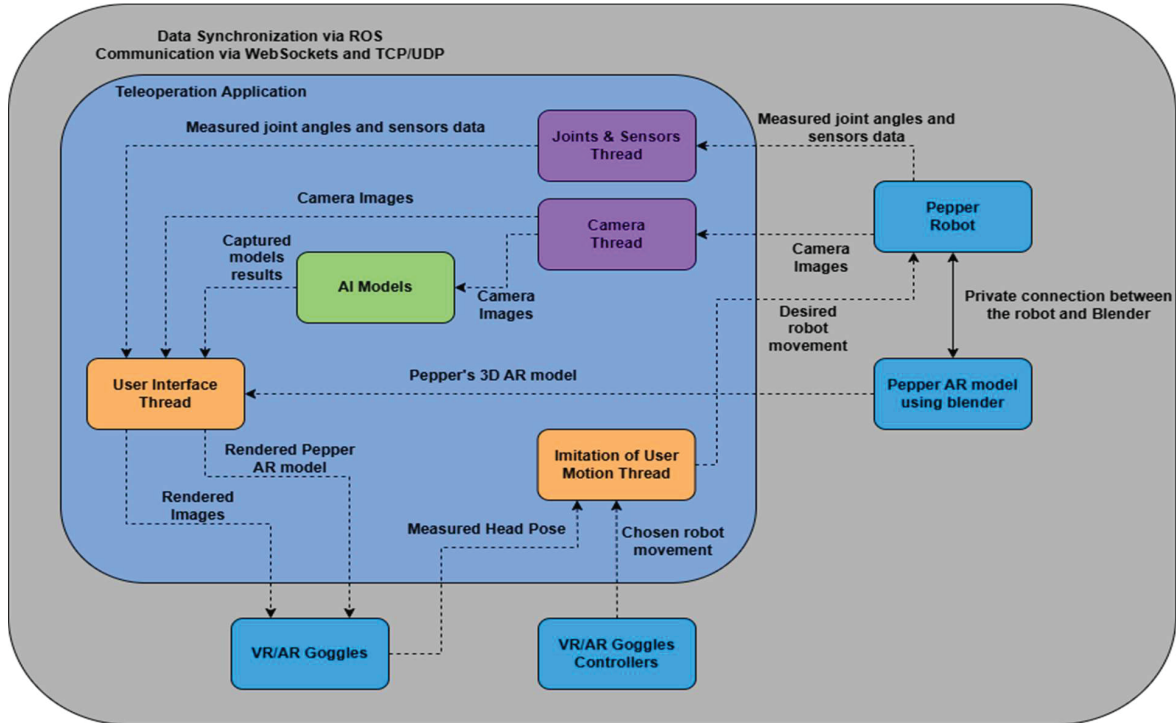


Fig. 1. Teleoperation framework architecture for Pepper robot.

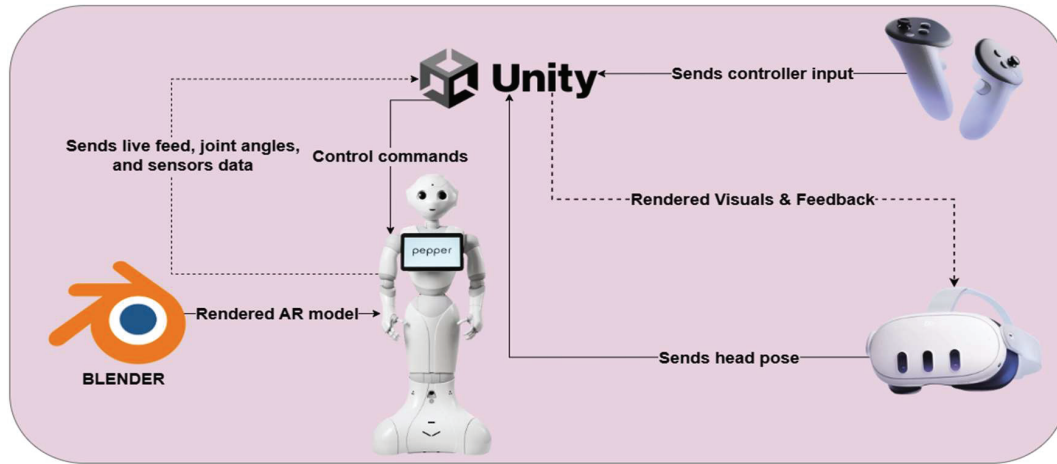


Fig. 2. Implementation and pipeline of the teleoperation process through the Meta Quest 3 goggles and Pepper robot.

transmission, a dual-protocol communication approach combining TCP and UDP connections was employed. Leveraging WebSockets for real-time, bidirectional communication between the operator and Pepper, allowing data to be sent to Pepper and received back from the Oculus simultaneously, ensuring seamless real-time feedback. TCP was designated for critical data, including control commands and sensor readings, ensuring reliable, loss-free transmission. This approach is essential for maintaining control integrity in precision-focused applications where missed data could compromise task accuracy. Concurrently, UDP was used to transmit the video feed from Pepper's camera to the VR goggles, enabling faster, low-latency streaming suited for real-time visual feedback, even if some packets are occasionally lost. This combined TCP/UDP configuration allows the operator to receive immediate visual updates while safeguarding the integrity of crucial control data, resulting in effective teleoperation performance.

B. Direct Control with VR

The system's VR integration, built using the Unity and Meta Quest3 framework, provides a fully immersive, intuitive interface for controlling Pepper, offering a level of precision and responsiveness that sets a new standard for human-robot interaction. As shown in Figure 2., the operator's head and hand movements are mapped directly to Pepper's corresponding actions in real time, eliminating the need for complex inverse kinematics calculations. This direct mapping enables smooth, fluid control, empowering the operator to perform delicate and precise maneuvers with ease, even in dynamic or high-pressure environments. Real-time sensor and camera feedback from Pepper are continuously streamed to the VR interface, providing the operator with a perspective that mirrors Pepper's own. Additionally, a study conducted with 20 participants aged 18–45, comprising 25% first-time users and 75% experienced users, demonstrated an average continuous usage time of 10–40 minutes without experiencing dizziness, highlighting the system's ergonomic and user-friendly design. This immersive setup allows the operator to perceive and respond to the environment as if physically present within Pepper's field of view, enhancing spatial awareness and task precision crucial for scenarios demanding fine motor skills and rapid responses.

C. AR Model Visualization

To further elevate situational awareness, the system leverages AR through the VR/AR Quest interface, presenting the operator with a dynamic, real-time 3D model of Pepper within their field of vision. As seen in Figure 2., this virtual model is meticulously designed in Blender to reflect Pepper's exact physical state, including joint angles, posture, and orientation, with real-time updates. This AR model serves as a powerful visualization tool, allowing operators to intuitively anticipate physical constraints and movement limitations. Continuous feedback on Pepper's positioning and joint movements enables proactive adjustments, which minimize the risk of errors and enhance control accuracy in complex operational settings. This enhancement fosters a deep sense of unity between the operator and Pepper, creating a level of intuitive understanding that is essential for high-stakes, precision-driven applications.

D. AI-Assisted Task Automation and Intelligent Feedback

AI-assisted features were integrated to reduce the cognitive load on the operator and improve efficiency. These features, which can be activated or deactivated based on the task requirements.

- The system integrates MiDaS for depth estimation, FaceNet for face recognition, a GPT-based knowledge model for contextual interactions, and DeepSpeech for speech recognition. All models utilize pre-trained versions and have performed effectively, enhancing Pepper's teleoperation capabilities across various scenarios [12,13,14,15].

E. Comprehensive Sensor Integration and Feedback

The teleoperation system fully utilizes Pepper's extensive multi-modal sensor suite, which includes depth cameras, temperature sensors, joint angle sensors, and battery status indicators, among others. Built on the NAOqi framework, all sensor data is transmitted in real time to the VR interface via WebSockets, utilizing both TCP and UDP connections, offering the operator a comprehensive, immediate overview of Pepper's physical condition and operational health. As illustrated in Figure 3., the sensor layout covers a wide range of inputs, providing the operator with detailed insights into Pepper's internal and external environment. These data streams enable the operator to

make preemptive adjustments, avoiding potential issues before they escalate. Additionally, built-in safety mechanisms allow Pepper to autonomously monitor critical sensor data, preventing hardware strain and enhancing operational safety and longevity. By ensuring the operator is fully aware of Pepper's physical state, the system maximizes control accuracy, reliability, and the robot's functional lifespan, making it a robust tool for demanding teleoperation tasks.

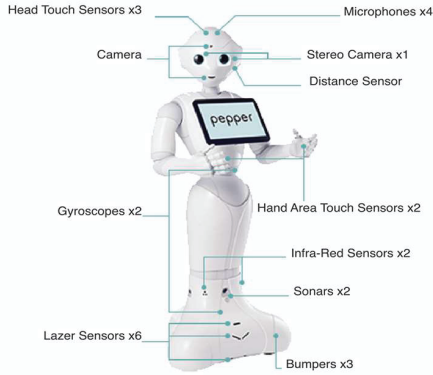


Fig. 3. Sensor and Hardware Components of the Pepper Robot.

V. RESULTS AND DISCUSSION

The subsequent sections provide a detailed analysis of the calculated latencies at each stage of the entire process flow illustrated in figure 2., including their mean, standard deviation, and mean squared error (MSE), which are displayed in the top right corner of all relevant figures.

A. Image Capture Latency

Figure 4. illustrates image capture latency, which depicts the time taken to capture each frame from the camera. Variability was observed across events, with peaks potentially attributable to hardware limitations or resource contention. The median capture latency hovers around 0.3 seconds, but the upper extremes exceed 0.5 seconds. Such peaks, particularly in real-time processing systems, could hinder continuous processing or increase overall task execution time. Consequently, reducing image capture latency could be beneficial for smoother, real-time applications.

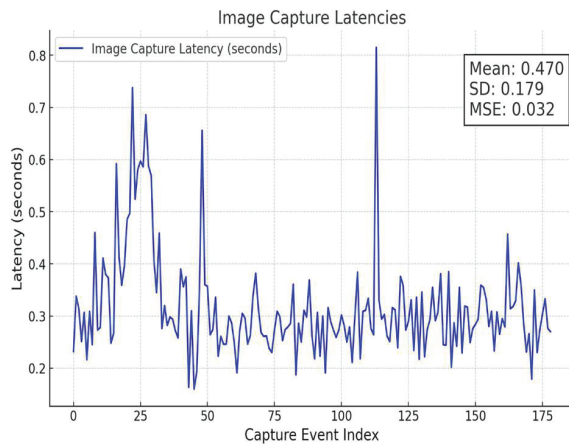


Fig. 4. Image Capture Latency

B. Image Processing Latency

As depicted in Figure 5., the image processing and display latency is predominantly near zero, with occasional instances reaching up to 0.001 seconds. This consistently low latency indicates that image processing has minimal impact on the system's total processing load. However, future optimization could address these sporadic increases, especially in systems sensitive to micro-latency changes. Overall, the stability of this metric suggests efficient image processing, which is vital for maintaining real-time performance.

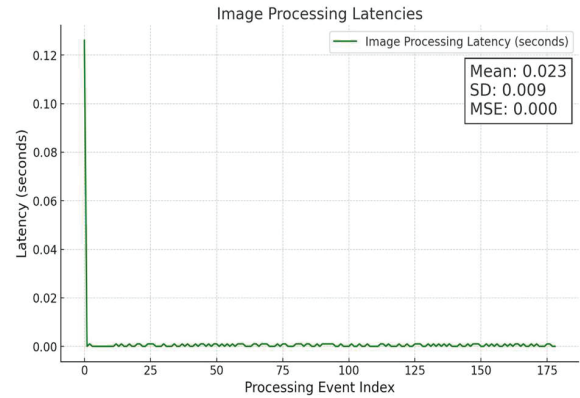


Fig. 5. Image processing Latency

C. Movement Latency

Figure 6. details the time taken for each of the forward, backward, left, and right movement commands executed within the robotic system. The average movement latency remains between 2.5 and 3.3 seconds, with forward and backward movements typically requiring slightly more time than lateral movements. This consistency is critical for ensuring predictable robotic behavior. However, the slight variability in latency across different movement commands may necessitate fine-tuning to balance command durations, ensuring uniform response times across all directions.

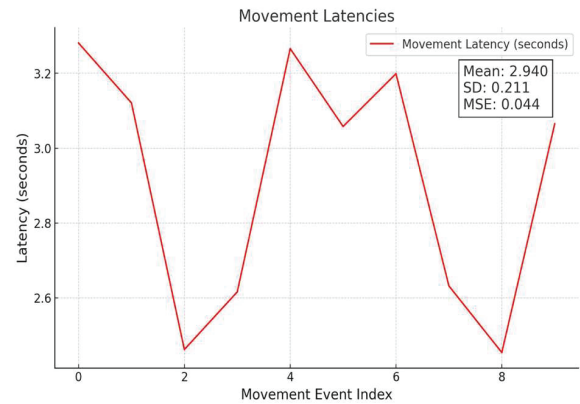


Fig. 6. Movement Latency

D. Overall Movement Cycle Latency

Figure 7. displays the latency across complete movement cycles, each encompassing a full process where the robot moves in all directions and returns to the starting point. As depicted in the inset diagram at the bottom left of figure 7., this includes the robot moving from the midpoint

to forward, forward back to the midpoint, midpoint to backward, backward to the midpoint, midpoint to right, right to the midpoint, midpoint to left, and left back to the midpoint. The cycle latency values range from approximately 28.0 to 28.2 seconds, showcasing remarkable stability across cycles. This uniformity in cycle times supports the system's reliability in executing movement commands with minimal deviation. For applications requiring strict time-sensitive operations, maintaining such consistency is essential.

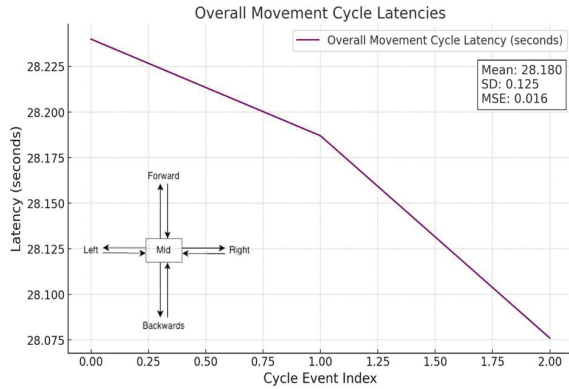


Fig. 7. Overall Movement Cycle Latency with an Inset graphical representation of the robot's movement.

The analysis of these latency graphs provides a comprehensive view of the system's performance. While image processing latency is minimal and stable, capturing and movement latencies present areas for optimization. Reducing image capture latency and balancing movement command latencies could yield a more responsive and efficient system, especially in applications demanding precise, real-time control. The stability observed in overall movement cycle latency underscores the system's potential reliability for consistent task execution. Additionally, the calculated MSE values demonstrated promising results, with values of 0.032, 0.0001, 0.044, and 0.016 for image processing, capturing, movement, and the overall movement cycle, respectively.

VI. CONCLUSION AND FUTURE WORK

This research introduces a cutting-edge teleoperation framework for the Pepper humanoid robot, seamlessly integrating VR, AR, and AI to achieve a new standard in responsive, intuitive control. By leveraging a dual WebSocket protocol that combines TCP and UDP, a robust, low-latency communication system was established, empowering operators to control Pepper with unparalleled accuracy in complex, high-stakes environments. VR immerses the operator in Pepper's perspective, while AR provides real-time updates on the robot's physical state, enhancing situational awareness and precision in every movement. The latency results confirm stable system performance, with image capture latency averaging 0.3 seconds, image processing near zero, movement latency ranging from 2.5 to 3.3 seconds across directions, and overall movement cycles consistently between 28.0 and 28.2 seconds. These metrics underscore the framework's precision and responsiveness in real-time applications.

Future work will focus on evaluating the system's scalability and performance under varying network

conditions. Furthermore, efforts will be directed toward developing custom AI models or fine-tuning the selected pretrained models to achieve optimized performance.

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